



BIPOLAR POWER SUPPLY With Adjustable Regulators

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LM317T and LM337T ICs are well-known, low-cost adjustable voltage regulators capable of delivering up to 1.5A current output with power dissipation of up to 20W. LM317T produces a positive output voltage, while LM337T produces a negative output voltage.

It is useful to have a bipolar power supply with LM317T and LM337T producing symmetrical output power. However, adjusting both output voltages simultaneously is an issue.

The usual solution is to build a tracking voltage regulator with operational amplifier, which tracks the positive or negative output of

the power supply. But here power supplies and other parameters of the operational amplifier can be limiting factors for the required power output.

Another solution is to use a stereo potentiometer (potmeter) to adjust both the output power supplies simultaneously. Good-quality stereo potentiometers have small differences (of about $\pm 5\%$) between the two outputs. If this difference is too large, you may use additional potentiometers to adjust the output voltages to exactly the same value.

Here we present the circuit of a bipolar power supply with LM317T

and LM337T adjustable regulators. The circuit provides the facility to adjust output voltages more precisely with individual potentiometers. Also, output voltages can be adjusted from near-ground levels, not from the typical $\pm 1.25V$.

Circuit and working

Circuit diagram of the bipolar power supply with LM317T and LM337T is shown in Fig. 1. It is built around an 18V-0-18V step-down transformer (X1), 1A bridge rectifier (BR1), variable positive-voltage regulator LM317T (IC1), variable negative-voltage regulator LM337T (IC2), eight 1N4001 diodes (D1 through

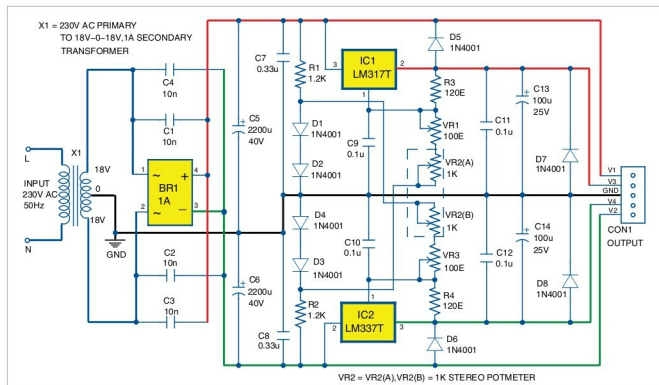


Fig. 1: Circuit of the bipolar power supply with adjustable regulators